

Dry Ice Inventory Optimization Report

Executive Summary

This report provides a comprehensive analysis of the dry ice inventory for the period from 02-Jul-2025 to 29-May-2026. The analysis identifies significant opportunities for cost savings and operational efficiency. By implementing an EOQ-based inventory policy, an estimated KSh 16,703 can be saved monthly, representing a 65.1% reduction in inventory costs.

Key Performance Indicators (KPIs)

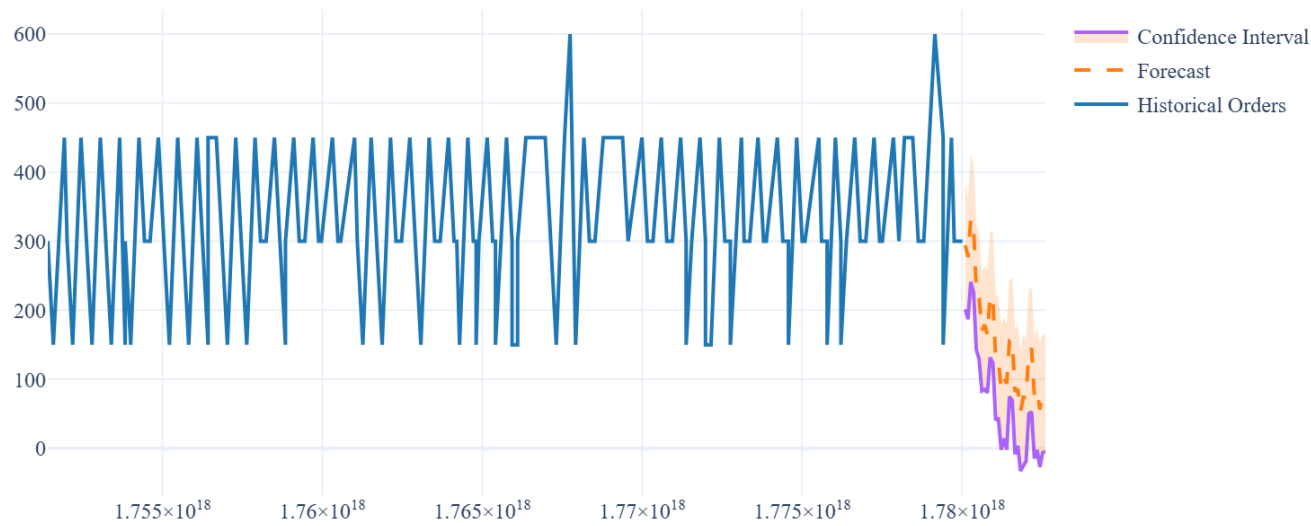
Potential Monthly Savings	KSh 16,703
Economic Order Quantity (EOQ)	2,754 kg
Recommended Safety Stock	169 kg
Reorder Point	2,923 kg
Average Monthly Demand	9,570 kg

Top Recommendations

- Adopt an EOQ-based ordering policy. Place orders for **2754 kg** at a time to minimize total inventory costs.
- Establish a safety stock level of **169 kg**. This buffer will protect against demand variability and prevent stockouts.
- Set a reorder point at **2923 kg**. A new order should be triggered automatically when inventory falls to this level.

Demand Forecast Analysis

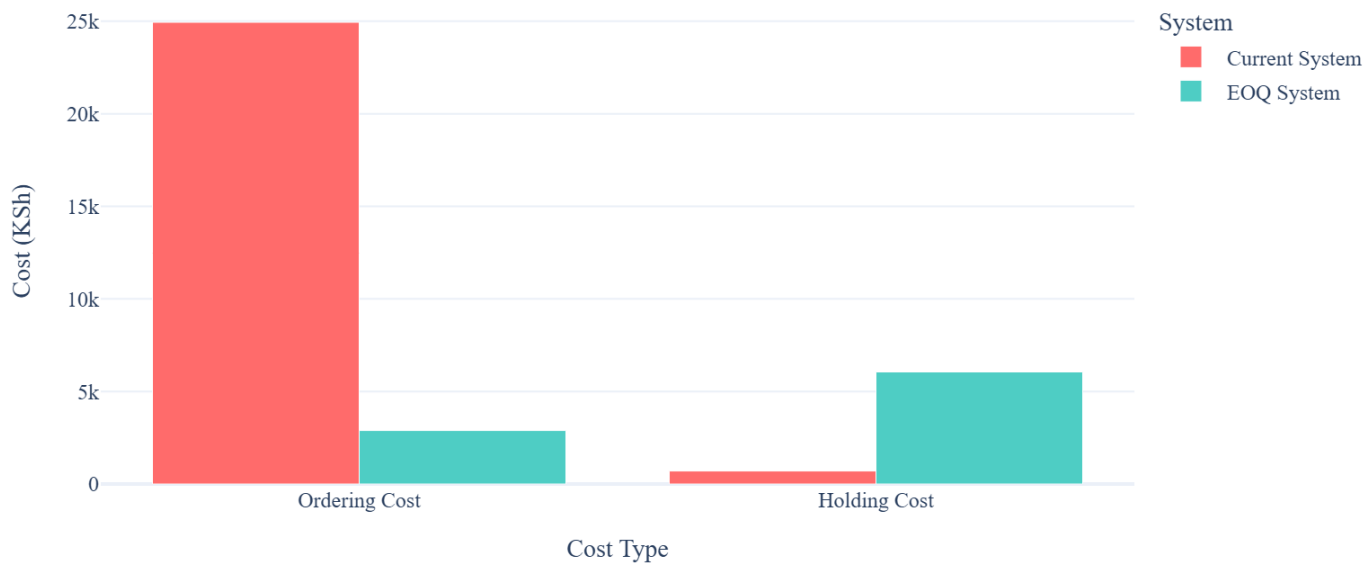
30-Day Demand Forecast



The 30-day demand forecast predicts fluctuating but consistent usage. The shaded area represents the 95% confidence interval, indicating the likely range of future demand. This forecast is crucial for planning orders and avoiding stockouts.

Cost Optimization Breakdown

Monthly Cost Comparison: Current vs. EOQ



The chart above clearly illustrates the cost benefits of switching to an EOQ-based system. While holding costs increase slightly due to larger average inventory, the reduction in ordering costs is substantial, leading to significant overall savings.